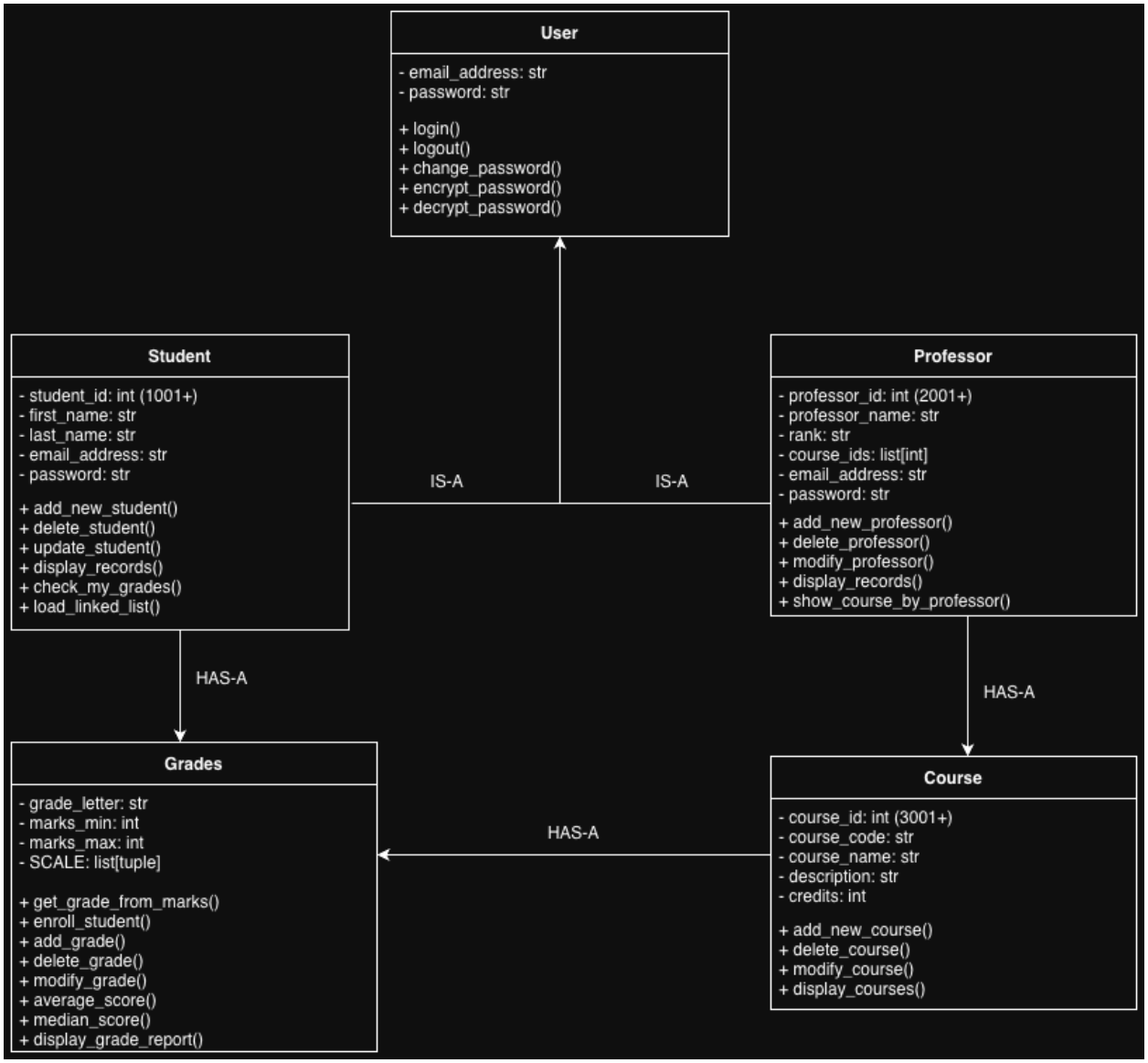
Github URL

https://github.com/popescoup/lab1-checkmygrade

Relationship Diagram



Unit Testing

```
lucapopescu@Lucas-MacBook-Pro Lab1_CheckMyGrade % python3 -m unittest test_checkmygrade -v
test_add_course (test_checkmygrade.TestCourseRecords.test_add_course)
Adding a course should: ... ok
test_delete_course (test_checkmygrade.TestCourseRecords.test_delete_course)
Deleting a course should: ... ok
test_modify_course (test_checkmygrade.TestCourseRecords.test_modify_course)
Modifying a course should: ... ok
test_add_professor (test_checkmygrade.TestProfessorRecords.test_add_professor)
Adding a professor should: ... ok
test_delete_professor (test_checkmygrade.TestProfessorRecords.test_delete_professor)
Deleting a professor should: ... ok
test_modify_professor (test_checkmygrade.TestProfessorRecords.test_modify_professor)
Modifying a professor should: ... ok
test_search_with_timing (test_checkmygrade.TestSearchAndSort.test_search_with_timing)
Search the LinkedList for a student by email address. ...
  [TIMING] Search (1 000 records, by email): 0.000020 seconds
ok
test_sort_by_email_asc (test_checkmygrade.TestSearchAndSort.test_sort_by_email_asc)
Sort 1 000 students by email address, ascending (A → Z). ...
  [TIMING] Sort by email ascending (1 000 records): 0.001667 seconds
ok
test_sort_by_email_desc (test_checkmygrade.TestSearchAndSort.test_sort_by_email_desc)
Sort 1 000 students by email address, descending (Z → A). ...
  [TIMING] Sort by email descending (1 000 records): 0.001410 seconds
ok
test_sort_by_marks_asc (test_checkmygrade.TestSearchAndSort.test_sort_by_marks_asc)
Sort 1 000 students by average marks, ascending. ...
  [TIMING] Sort by marks ascending (1 000 records): 0.001615 seconds
ok
test_sort_by_marks_desc (test_checkmygrade.TestSearchAndSort.test_sort_by_marks_desc)
Sort 1 000 students by average marks, descending. ...
  [TIMING] Sort by marks descending (1 000 records): 0.001591 seconds
ok
test_add_student (test_checkmygrade.TestStudentRecords.test_add_student)
Adding a student should: ... ok
test_bulk_add_1000_students (test_checkmygrade.TestStudentRecords.test_bulk_add_1000_students)
Bulk-insert 1 000 student records and verify: ...
  [TIMING] Bulk insert 1000 students: 2.1175 seconds
ok
test_delete_student (test_checkmygrade.TestStudentRecords.test_delete_student)
Deleting a student should: ... ok
test_modify_student (test_checkmygrade.TestStudentRecords.test_modify_student)
Modifying a student should: ... ok

----------------------------------------------------------------------
Ran 15 tests in 14.365s

OK
```

Functionality Testing

Registering student

```
------------------------------------------------------------
Welcome to CheckMyGrade
------------------------------------------------------------
1. Login
2. Register (students only)
3. Exit
------------------------------------------------------------
Select option: 2

------------------------------------------------------------
Student Registration
------------------------------------------------------------
Email address: bob@sjsu.edu
First name: Bob
Last name: Guy
Password: BobIsCool1
Confirm password: BobIsCool1

Registration successful! Your student ID is 1001.
You can now log in.
```

```
------------------------------------------------------------
Welcome to CheckMyGrade
------------------------------------------------------------
1. Login
2. Register (students only)
3. Exit
------------------------------------------------------------
Select option: 1

------------------------------------------------------------
Login
------------------------------------------------------------
Email: bob@sjsu.edu
Password: BobIsCool1

Welcome back! Logged in as student.
```

Enrolling & basic functionality

```
------------------------------------------------------------
Student Menu  [bob@sjsu.edu]
------------------------------------------------------------
1. View my grades
2. View all courses
3. Enroll in a course
4. Unenroll from a course
5. Change password
6. Logout
------------------------------------------------------------
Select option: 3

------------------------------------------------------------
Enroll in a Course
------------------------------------------------------------

ID       Code         Name                      Credits   Description
-------- ------------ ------------------------- --------- ------------------------------------
3001     DATA200      Programming for Data Analytics 3         Basics of Python Programming for numerical implementation of mathematical and statistical algorithms
3002     DATA201      Database Technologies     3         Fundamentals of SQL and database design

Enter Course ID to enroll in: 3001

Enrolled successfully.
```

```
-------------------------------------------------------
Student Menu  [bob@sjsu.edu]
-------------------------------------------------------
1. View my grades
2. View all courses
3. Enroll in a course
4. Unenroll from a course
5. Change password
6. Logout
-------------------------------------------------------
Select option: 2

-------------------------------------------------------
All Courses
-------------------------------------------------------

ID       Code         Name                      Credits   Description
-------- ------------ ------------------------- --------- -----------------------------------
3001     DATA200      Programming for Data Analytics 3         Basics of Python Programming for numerical implementation of mathematical and statistical algorithms
3002     DATA201      Database Technologies     3         Fundamentals of SQL and database design
```

```
-------------------------------------------------------
Student Menu  [bob@sjsu.edu]
-------------------------------------------------------
1. View my grades
2. View all courses
3. Enroll in a course
4. Unenroll from a course
5. Change password
6. Logout
-------------------------------------------------------
Select option: 4

-------------------------------------------------------
Unenroll from a Course
-------------------------------------------------------

Course Code    Course Name                Marks    Grade
-------------- -------------------------- -------- ------
DATA200        Programming for Data Analytics          N/A
DATA201        Database Technologies              N/A

Enter Course ID to unenroll from: 3002

Unenrolled successfully.
```

```
-------------------------------------------------------
Student Menu  [bob@sjsu.edu]
-------------------------------------------------------
1. View my grades
2. View all courses
3. Enroll in a course
4. Unenroll from a course
5. Change password
6. Logout
-------------------------------------------------------
Select option: 1

-------------------------------------------------------
My Grades
-------------------------------------------------------

Course Code    Course Name                Marks    Grade
-------------- -------------------------- -------- ------
DATA200        Programming for Data Analytics          N/A
```

Changing password

```
------------------------------------------------------------
Student Menu  [bob@sjsu.edu]
------------------------------------------------------------
1. View my grades
2. View all courses
3. Enroll in a course
4. Unenroll from a course
5. Change password
6. Logout
------------------------------------------------------------
Select option: 5

------------------------------------------------------------
Change Password
------------------------------------------------------------
Current password: BobIsCool1
New password: BobIsCool2
Confirm new password: BobIsCool2

Password changed successfully.
```

```
------------------------------------------------------------
Student Menu  [bob@sjsu.edu]
------------------------------------------------------------
1. View my grades
2. View all courses
3. Enroll in a course
4. Unenroll from a course
5. Change password
6. Logout
------------------------------------------------------------
Select option: 6

Logged out successfully.
```

Demonstrating successful password change

```
------------------------------------------------------------
Welcome to CheckMyGrade
------------------------------------------------------------
1. Login
2. Register (students only)
3. Exit
------------------------------------------------------------
Select option: 1

------------------------------------------------------------
Login
------------------------------------------------------------
Email: bob@sjsu.edu
Password: BobIsCool2

Welcome back! Logged in as student.
```

Professor login

```
------------------------------------------------------------
Welcome to CheckMyGrade
------------------------------------------------------------
1. Login
2. Register (students only)
3. Exit
------------------------------------------------------------
Select option: 1

------------------------------------------------------------
Login
------------------------------------------------------------
Email: luca@sjsu.edu
Password: LucaIsCool1

Welcome back! Logged in as professor.
```

Showcasing student management functionalities

```
-------------------------------------------------------
Professor Menu  [luca@sjsu.edu]
-------------------------------------------------------
1. Student Management
2. Course Management
3. Professor Management
4. Enrollment & Grades
5. Reports & Statistics
6. Change Password
7. Logout
-------------------------------------------------------
Select option: 1

-------------------------------------------------------
Student Management
-------------------------------------------------------
1. Add student
2. Delete student
3. Modify student
4. Search student
5. Sort & display students
6. Display all students
0. Back
-------------------------------------------------------
Select option: 6

-------------------------------------------------------
All Students
-------------------------------------------------------

ID       First Name      Last Name       Email
-------- --------------- --------------- ------------------------------
1002     John            Dude            john@sjsu.edu
1001     Bob             Guy             bob@sjsu.edu
```

```
-------------------------------------------------------
Student Management
-------------------------------------------------------
1. Add student
2. Delete student
3. Modify student
4. Search student
5. Sort & display students
6. Display all students
0. Back
-------------------------------------------------------
Select option: 1

-------------------------------------------------------
Add Student
-------------------------------------------------------
Email address: dude@sjsu.edu
First name: Dude
Last name: Guy
Temporary password: DudeIsGuy

Student added. ID: 1003
```

```
--------------------------------------------------------
Student Management
--------------------------------------------------------
1. Add student
2. Delete student
3. Modify student
4. Search student
5. Sort & display students
6. Display all students
0. Back
--------------------------------------------------------
Select option: 6

--------------------------------------------------------
All Students
--------------------------------------------------------

ID       First Name      Last Name       Email
-------- --------------- --------------- ------------------------------
1002     John            Dude            john@sjsu.edu
1003     Dude            Guy             dude@sjsu.edu
1001     Bob             Guy             bob@sjsu.edu
```

```
--------------------------------------------------------
Student Management
--------------------------------------------------------
1. Add student
2. Delete student
3. Modify student
4. Search student
5. Sort & display students
6. Display all students
0. Back
--------------------------------------------------------
Select option: 2

--------------------------------------------------------
Delete Student
--------------------------------------------------------
Student ID: 1003
Delete student 1003? This also removes their enrollments (y/n): y

Student deleted.
```

```
------------------------------------------------------------
Student Management
------------------------------------------------------------
1. Add student
2. Delete student
3. Modify student
4. Search student
5. Sort & display students
6. Display all students
0. Back
------------------------------------------------------------
Select option: 3

------------------------------------------------------------
Modify Student
------------------------------------------------------------
Student ID: 1001
Leave a field blank to keep the current value.
New first name (or blank): Bobby
New last name (or blank):
New email (or blank):

Student updated.
```

```
------------------------------------------------------------
Student Management
------------------------------------------------------------
1. Add student
2. Delete student
3. Modify student
4. Search student
5. Sort & display students
6. Display all students
0. Back
------------------------------------------------------------
Select option: 4

------------------------------------------------------------
Search Student
------------------------------------------------------------
Search by:
1. Student ID
2. Email address
Select option: 1
Student ID: 1001

Search completed in 0.000042 seconds.

ID       : 1001
Name     : Bobby Guy
Email    : bob@sjsu.edu
```

Student search and sorting

```
------------------------------------------------------------
Student Management
------------------------------------------------------------
1. Add student
2. Delete student
3. Modify student
4. Search student
5. Sort & display students
6. Display all students
0. Back
------------------------------------------------------------
Select option: 4

------------------------------------------------------------
Search Student
------------------------------------------------------------
Search by:
1. Student ID
2. Email address
Select option: 2
Email address: john@sjsu.edu

Search completed in 0.000013 seconds.

ID       : 1002
Name     : John Dude
Email    : john@sjsu.edu
```

```
------------------------------------------------------------
Professor Menu  [luca@sjsu.edu]
------------------------------------------------------------
1. Student Management
2. Course Management
3. Professor Management
4. Enrollment & Grades
5. Reports & Statistics
6. Change Password
7. Logout
------------------------------------------------------------
Select option: 4

------------------------------------------------------------
Enrollment & Grade Management
------------------------------------------------------------
1. Assign / update grade
2. Delete enrollment
0. Back
------------------------------------------------------------
Select option: 1

------------------------------------------------------------
Assign / Update Grade
------------------------------------------------------------
Student ID: 1001
Course ID: 3001
Marks (0-100): 100

Grade assigned: 100 → A+
```

```
------------------------------------------------------------
Student Management
------------------------------------------------------------
1. Add student
2. Delete student
3. Modify student
4. Search student
5. Sort & display students
6. Display all students
0. Back
------------------------------------------------------------
Select option: 5

------------------------------------------------------------
Sort & Display Students
------------------------------------------------------------
Sort by:
1. Last name (A → Z)
2. Last name (Z → A)
3. Marks (low → high)
4. Marks (high → low)
5. Email (A → Z)
6. Email (Z → A)
Select option: 1

Sort completed in 0.000017 seconds.

ID       First Name      Last Name       Email
------------------------------------------------------------
1002     John            Dude            john@sjsu.edu
1001     Bobby           Guy             bob@sjsu.edu
```

```
------------------------------------------------------------
Sort & Display Students
------------------------------------------------------------
Sort by:
1. Last name (A → Z)
2. Last name (Z → A)
3. Marks (low → high)
4. Marks (high → low)
5. Email (A → Z)
6. Email (Z → A)
Select option: 3

Sort completed in 0.000009 seconds.

ID       First Name      Last Name       Avg Marks  Email
------------------------------------------------------------
1002     John            Dude            95.0       john@sjsu.edu
1001     Bobby           Guy             100.0      bob@sjsu.edu
```

```
------------------------------------------------------------
Sort & Display Students
------------------------------------------------------------
Sort by:
1. Last name (A → Z)
2. Last name (Z → A)
3. Marks (low → high)
4. Marks (high → low)
5. Email (A → Z)
6. Email (Z → A)
Select option: 6

Sort completed in 0.000017 seconds.

ID       First Name      Last Name       Email
------------------------------------------------------------
1002     John            Dude            john@sjsu.edu
1001     Bobby           Guy             bob@sjsu.edu
```

Showcasing course management functionalities

```
-----------------------------------------------------
Professor Menu  [luca@sjsu.edu]
-----------------------------------------------------
1. Student Management
2. Course Management
3. Professor Management
4. Enrollment & Grades
5. Reports & Statistics
6. Change Password
7. Logout
-----------------------------------------------------
Select option: 2

-----------------------------------------------------
Course Management
-----------------------------------------------------
1. Add course
2. Delete course
3. Modify course
4. Display all courses
0. Back
-----------------------------------------------------
Select option: 4

-----------------------------------------------------
All Courses
-----------------------------------------------------

ID       Code         Name                     Credits   Description
-------- ------------ ------------------------ --------- -----------------------------------
3001     DATA200      Programming for Data Analytics 3         Basics of Python Programming for numerical implementation of mathematical and statistical algorithms
3002     DATA201      Database Technologies     3         Fundamentals of SQL and database design
```

```
------------------------------------------------------------
Course Management
------------------------------------------------------------
1. Add course
2. Delete course
3. Modify course
4. Display all courses
0. Back
------------------------------------------------------------
Select option: 1

------------------------------------------------------------
Add Course
------------------------------------------------------------
Course code (e.g. DATA200): DATA202
Course name: Math for Data Science
Description: Stats, calc, linear algebra
Credits: 3

Course added. ID: 3003
```

```
------------------------------------------------------------
Course Management
------------------------------------------------------------
1. Add course
2. Delete course
3. Modify course
4. Display all courses
0. Back
------------------------------------------------------------
Select option: 2

------------------------------------------------------------
Delete Course
------------------------------------------------------------
Course ID: 3002
Delete course 3002? (y/n): y

Course deleted.
```

```
------------------------------------------------------------
Course Management
------------------------------------------------------------
1. Add course
2. Delete course
3. Modify course
4. Display all courses
0. Back
------------------------------------------------------------
Select option: 3

------------------------------------------------------------
Modify Course
------------------------------------------------------------
Course ID: 3003
Leave a field blank to keep the current value.
New course code (or blank):
New course name (or blank): Math for Data Analytics
New description (or blank):
New credits (or blank): 5

Course updated.
```

```
-----------------------------------------------------
Course Management
-----------------------------------------------------
1. Add course
2. Delete course
3. Modify course
4. Display all courses
0. Back
-----------------------------------------------------
Select option: 4

-----------------------------------------------------
All Courses
-----------------------------------------------------

ID       Code         Name                      Credits   Description
------- ----------- ------------------------- --------- -----------------------------------
3001     DATA200      Programming for Data Analytics 3         Basics of Python Programming for numerical implementation of mathematical and statistical algorithms
3003     DATA202      Math for Data Analytics   5         Stats, calc, linear algebra
```

Professor report generation functionality

```
------------------------------------------------------------
Professor Menu  [luca@sjsu.edu]
------------------------------------------------------------
1. Student Management
2. Course Management
3. Professor Management
4. Enrollment & Grades
5. Reports & Statistics
6. Change Password
7. Logout
------------------------------------------------------------
Select option: 5

------------------------------------------------------------
Reports & Statistics
------------------------------------------------------------
1. Grade report by course
2. Grade report by professor
3. Grade report by student
4. Average score by course
5. Median score by course
0. Back
------------------------------------------------------------
Select option: 1

------------------------------------------------------------
Grade Report — By Course
------------------------------------------------------------
Course ID: 3001

Student ID   Student Name           Course Code    Marks    Grade
------------ ---------------------- -------------- -------- ------
1001         Bobby Guy              DATA200        100      A+
1002         John Dude              DATA200        95       A
```

```
------------------------------------------------------------
Reports & Statistics
------------------------------------------------------------
1. Grade report by course
2. Grade report by professor
3. Grade report by student
4. Average score by course
5. Median score by course
0. Back
------------------------------------------------------------
Select option: 2

------------------------------------------------------------
Grade Report — By Professor
------------------------------------------------------------
Professor ID: 2001

Student ID   Student Name           Course Code    Marks    Grade
------------ ---------------------- -------------- -------- ------
1001         Bobby Guy              DATA200        100      A+
1002         John Dude              DATA200        95       A
```

```
------------------------------------------------------------
Reports & Statistics
------------------------------------------------------------
1. Grade report by course
2. Grade report by professor
3. Grade report by student
4. Average score by course
5. Median score by course
0. Back
------------------------------------------------------------
Select option: 3

------------------------------------------------------------
Grade Report — By Student
------------------------------------------------------------
Student ID: 1002

Student ID   Student Name           Course Code    Marks    Grade
------------ ---------------------- -------------- -------- ------
1002         John Dude              DATA200        95       A
```

```
------------------------------------------------------------
Reports & Statistics
------------------------------------------------------------
1. Grade report by course
2. Grade report by professor
3. Grade report by student
4. Average score by course
5. Median score by course
0. Back
------------------------------------------------------------
Select option: 4

------------------------------------------------------------
Average Score — By Course
------------------------------------------------------------
Course ID: 3001

Average score: 97.5
```

```
------------------------------------------------------------
Reports & Statistics
------------------------------------------------------------
1. Grade report by course
2. Grade report by professor
3. Grade report by student
4. Average score by course
5. Median score by course
0. Back
------------------------------------------------------------
Select option: 5

------------------------------------------------------------
Median Score — By Course
------------------------------------------------------------
Course ID: 3001

Median score: 97.5
```

Professor management functionalities

```
-------------------------------------------------------
Professor Management
-------------------------------------------------------
1. Add professor
2. Delete professor
3. Modify professor
4. Display all professors
5. View my courses
0. Back
-------------------------------------------------------
Select option: 4

-------------------------------------------------------
All Professors
-------------------------------------------------------

ID       Name                     Rank                 Email                        Courses
-------- ------------------------ -------------------- ---------------------------- ---------------
2001     Luca Popescu             Senior Professor     luca@sjsu.edu                3001,3002
```

```
-------------------------------------------------------
Professor Management
-------------------------------------------------------
1. Add professor
2. Delete professor
3. Modify professor
4. Display all professors
5. View my courses
0. Back
-------------------------------------------------------
Select option: 5

-------------------------------------------------------
My Courses
-------------------------------------------------------

Courses taught by Luca Popescu:

ID       Code         Name                       Credits
-------- ------------ -------------------------- -------
3001     DATA200      Programming for Data Analytics 3
```

```
------------------------------------------------------------
Professor Management
------------------------------------------------------------
1. Add professor
2. Delete professor
3. Modify professor
4. Display all professors
5. View my courses
0. Back
------------------------------------------------------------
Select option: 1

------------------------------------------------------------
Add Professor
------------------------------------------------------------
Email address: prof@sjsu.edu
Professor name: Prof
Rank (e.g. Senior Professor, Lecturer): Lecturer
Temporary password: ProfIsCool1

Professor added. ID: 2002
```

```
------------------------------------------------------------
Welcome to CheckMyGrade
------------------------------------------------------------
1. Login
2. Register (students only)
3. Exit
------------------------------------------------------------
Select option: 1

------------------------------------------------------------
Login
------------------------------------------------------------
Email: prof@sjsu.edu
Password: ProfIsCool1

Welcome back! Logged in as professor.
```

```
------------------------------------------------------------
Professor Menu  [prof@sjsu.edu]
------------------------------------------------------------
1. Student Management
2. Course Management
3. Professor Management
4. Enrollment & Grades
5. Reports & Statistics
6. Change Password
7. Logout
------------------------------------------------------------
Select option: 7

Logged out successfully.

------------------------------------------------------------
Welcome to CheckMyGrade
------------------------------------------------------------
1. Login
2. Register (students only)
3. Exit
------------------------------------------------------------
Select option: 3

Goodbye!
```